

RENO COMPASS BLUEPRINT PLAN

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1. Project Summary

Project: Hall/Guest Bathroom Remodel (bathroom_remodel)

Goal: Bring hall/guest bathroom in line with rest of the house, which was renovated earlier. Replace aged vanity, add tub glass separator, fix shower height, add mirror, add IKEA standalone cabinet storage. Full gut removal but no changes to plumbing, walls, electricity positions. Re-drywall, install porcelain tile flooring, new light fixtures, new exhaust fan, and paint.

Property: Zip 95120, independent house, 50 sqft

Chosen design: preferred (est. \$25,000-\$28,000)

2. Safety & Permit (review this first)

- New Shower Valve: TIER 2 PERMITTED (source: IRC P2708.4, RD1-G2 (plumbing modification))
- New Exhaust Fan: TIER 2 PERMITTED (source: IRC R303.3, M1507, RD1-D1, RD1-D3, RD1-G4 (mechanical permit))
- Occupancy/Vacancy Sensor for Lighting: TIER 2 PERMITTED (source: RD1-G1 (electrical modification))
- Demolition (Gut Removal), Re-drywalling, and Painting: TIER 3 PROCEED (source: Agent Classification)
- New Porcelain Tile Flooring: TIER 3 PROCEED (source: Agent Classification)
- New Vanity, Countertop, and Undermount Sink: TIER 3 PROCEED (source: Agent Classification)
- New Toilet: TIER 3 PROCEED (source: Agent Classification)
- Tub Glass Separator: TIER 3 PROCEED (source: Agent Classification)
- Shower Niche: TIER 3 PROCEED (source: Agent Classification)
- New Light Fixtures: TIER 3 PROCEED (source: Agent Classification)
- Wall-Mounted Mirror, Freestanding Cabinet, and Wall Hanger: TIER 3 PROCEED (source: Agent Classification)

Permits are required. Confirm the full scope with your local AHJ.

3. Budget

Target: \$20,000 Ceiling: \$25,000

Ballpark: \$13,950-\$21,700 (+ contingency \$2,162-\$3,364)

Reality check: plausible

All costs are estimated ranges - verify locally with real quotes.

4. Logistics & Displacement

Can live through it: True

Feasibility verdict: proceed

6. Your DIY Plan

- New Shower Valve (tier 2 permitted)

1. Obtain a Plumbing Permit (Permit Hold-Point 1): Before you begin any work, apply for and obtain the necessary plumbing permit from your local building department. They will provide guidelines for inspections.

2. Prepare the Area and Shut Off Water:

Clear the area around the shower/tub.

Crucially, turn off the main water supply to your house. Open a lower-level faucet to drain the water lines.

Protect the tub/shower basin from debris or damage. You may need to remove existing wall coverings (like drywall or tile) to access the old valve, which you have already planned to do as part of your gut renovation.

3. Access and Remove the Old Valve:

If the wall is not already open from demolition, carefully cut open the wall around the existing valve to gain full access to the plumbing connections.

Use your pipe cutter to carefully cut the hot and cold water supply lines, and the lines going to the showerhead and tub spout (if applicable). Have a bucket ready for any remaining water.

Remove the old valve from its mounting.

4. Mount the New Shower Valve:

Position the new valve body. Ensure it is level and at the correct height for your rain shower system (typically higher than a standard showerhead). The valve should be securely fastened to the wall studs or a wooden cross-brace.

Follow the manufacturer's instructions for the specific rough-in depth, ensuring it will be flush with your finished wall surface (considering the thickness of cement board and tile).

5. Connect New Plumbing Lines:

Connect the hot and cold water supply lines to the corresponding ports on the new valve. Ensure these connections are secure and watertight, using thread seal tape or pipe dope on threaded connections.

Connect the lines from the valve to your showerhead and tub spout (if you have one). Again, ensure all connections are secure.

6. Rough-In Plumbing Inspection (Permit Hold-Point 2): Once the new valve is installed and all plumbing connections are made before you close up the wall, you must schedule and pass a rough-in plumbing inspection. The inspector will check the connections, valve installation, and ensure it meets code.

7. Water Pressure Test: After passing inspection and before closing the wall, turn the main water supply back on (slowly) and carefully check for any leaks around all new connections. This is a critical step to avoid future water damage.

8. Install Wall Coverings and Trim:

Once satisfied with the leak test and after the rough-in inspection is approved, you can proceed with installing your cement backer board and tiling.

After the wall is finished, install the decorative trim, handle, and showerhead/tub spout according to the manufacturer's instructions.

9. Final Plumbing Inspection (Permit Hold-Point 3): After the entire shower system is installed and operational, you must schedule and pass a final plumbing inspection. The inspector will verify the complete installation and proper function of the valve.

Hold point: Permit Hold-Point 1

Hold point: Permit Hold-Point 2

Hold point: Permit Hold-Point 3

Tool: Adjustable Wrench and/or Pipe Wrench - For connecting and tightening plumbing fittings.

Tool: Screwdriver Set - To remove and install the valve trim and mounting screws.

Tool: PEX Crimping Tool and Cutter - For making secure connections (if using PEX tubing). (if using PEX tubing)

Tool: Propane Torch, Solder, Flux, and Deburring Tool - For traditional copper pipe connections (if using Copper piping). (if using Copper piping)

Tool: Pipe Cutter - For cleanly cutting existing water lines and new piping.

Tool: Tape Measure and Level - For accurate positioning and ensuring the valve is plumb.

Tool: Stud Finder - To locate studs for safe cutting (if opening wall further).

Tool: Bucket and Rags - To catch any residual water when disconnecting old lines.

Tool: Thread Seal Tape (Teflon Tape) or Pipe Dope - To ensure watertight threaded connections.

Tool: Safety Glasses and Gloves - Appropriate personal protective equipment.

- Occupancy/Vacancy Sensor for Lighting (tier 2 permitted)

- Demolition (Gut Removal), Re-drywalling, and Painting (tier 3 proceed)

1. Phase 1: Demolition (Gut Removal)

2. Preparation and Safety:

Crucially, turn off the main water supply to the bathroom or the entire house. Open a lower-level faucet to drain the lines.

Turn off the electrical breaker(s) to the bathroom circuit(s). Use a non-contact voltage tester to confirm power is off at switches and outlets.

Wear all appropriate Personal Protective Equipment (PPE): heavy-duty gloves, safety glasses, dust mask/respirator, and sturdy shoes.

Cover vents and seal the bathroom doorway with plastic sheeting and painter's tape to contain dust.

Remove anything from the room you want to keep.

3. Fixture Removal:

Toilet: Turn off the toilet's supply valve, flush, disconnect the supply line, unbolt from the floor, and remove. Plug the waste opening.

Vanity/Sink: Turn off water supply, disconnect supply lines and P-trap (have a bucket ready), unfasten from the wall, and remove the countertop/sink and then the cabinet.

Mirror/Other Accessories: Carefully remove any wall-mounted mirrors, cabinets, light fixtures (after confirming power is off), and other accessories.

Tub/Shower: Remove any glass enclosures or doors. If removing the tub itself, this can be heavy and may require assistance or breaking it into pieces (especially cast iron).

4. Wall and Floor Removal:

Drywall/Tile Backer: Use a utility knife to score the perimeter of the drywall/backer board. Use a pry bar or hammer to break and remove sections. Remove screws/nails as you go. For tiled walls, you may need a demolition hammer for thick mortar beds.

Flooring: For tile, use a hammer and chisel or a pry bar to lift and break up tiles, working down to the subfloor. Remove any underlayment or mortar bed.

5. Initial Cleanup:

Collect all demolition debris into heavy-duty trash bags or a dumpster.

Sweep and vacuum the area thoroughly.

Inspect studs, subfloor, and ceiling joists for any damage, moisture, or necessary repairs. This is the time to address any structural concerns before rebuilding.

6. Phase 2: Re-drywalling

7. Wall Preparation:

Ensure studs are clean, dry, and free of protruding nails/screws.

Install new blocking or framing as needed for the vanity, shower niche, wall hanger, and any other fixtures requiring solid backing.

8. Install Drywall:

Measure and cut drywall sheets to fit, using a utility knife to score and snap. Use a drywall saw for cutting around outlets and pipes.

Secure drywall sheets to studs using drywall screws, spacing them appropriately (typically every 6-8 inches on edges and 12 inches in the field). Ensure screws are slightly recessed but do not break the paper face.

For areas that will be tiled (like the tub surround), install cement backer board according to the manufacturer's instructions.

9. Tape and Mud Joints:

Apply joint tape (paper or fiberglass mesh) over all seams between drywall sheets.

Using a 6-inch drywall knife, apply a thin, even coat of joint compound (mud) over the tape, filling the seam. Also, apply mud to all screw heads.

Allow the first coat to dry completely (check product instructions, usually several hours or overnight).

10. Second and Third Coats:

Apply a second, wider coat of joint compound using a 10-inch knife, feathering the edges out from the seam to blend with the wall surface.

Once dry, apply a third, even wider and thinner coat using a 12-inch knife for a smooth, seamless finish.

Allow each coat to dry completely before applying the next.

11. Sanding:

Once all joint compound is thoroughly dry, lightly sand all mudded areas with a sanding sponge or pole sander until smooth and flat. Wear a dust mask/respirator!

Wipe down walls to remove all dust before priming.

12. Phase 3: Painting

13. Preparation:

Clean all wall surfaces thoroughly to remove any dust, grime, or sanding residue.

Apply painter's tape to protect trim, ceilings, and any fixtures you've already installed or don't want painted.

Lay down drop cloths to protect the floor and any remaining fixtures.

Fill any small imperfections or nail holes with spackle, sand smooth once dry, and wipe clean.

14. Prime:

Apply a high-quality primer to all new drywall and patched areas. Primer helps achieve a uniform finish and improves paint adhesion.

Allow primer to dry completely according to manufacturer instructions.

15. Paint:

Start by "cutting in" ? using a paintbrush to paint along the edges, corners, and around taped-off areas.

Use a roller to paint the larger wall and ceiling surfaces, applying even coats. Work in small sections, moving from top to bottom.

Allow the first coat to dry completely (check manufacturer's recommendations for recoat times).

Apply a second coat of paint, repeating the cutting-in and rolling process. More coats may be needed for a deeper, more uniform color.

Once the final coat is applied and still wet (or slightly tacky), carefully remove the painter's tape for the cleanest lines.

16. Final Cleanup:

Clean brushes and rollers thoroughly, or dispose of them properly.

Remove all drop cloths and dispose of any remaining paint supplies responsibly.

Tool: Heavy-Duty Gloves - To protect hands during all phases of work.

Tool: Safety Glasses/Goggles - Essential eye protection for demolition, cutting, and sanding.

Tool: Dust Mask/Respirator - Critical for protecting your lungs from dust, especially during demolition, sanding drywall, and painting.

Tool: Sturdy Closed-Toe Shoes/Work Boots - To protect your feet.

Tool: Pry Bar Set - For removing trim, baseboards, and breaking up drywall/tile backer.

Tool: Claw Hammer - General utility.

Tool: Utility Knife - For scoring drywall, cutting various materials, and opening packaging.

Tool: Large Buckets/Trash Cans & Heavy-Duty Trash Bags - For debris collection.

Tool: Shovel and Broom - For cleanup.

Tool: Wheelbarrow or Heavy-Duty Dolly - For moving heavier debris like old toilets, vanities, or large sections of tile. (Optional - Rent/Buy)

Tool: Demolition Hammer / Jackhammer - Highly recommended if your old tile is set in a thick mortar bed, it will save significant time and effort. (Optional - Rent)

Tool: Drill with Drywall Bit Attachment - For efficiently securing drywall screws without over-penetrating.

Tool: T-Square or Drywall Square - For accurately measuring and marking straight cuts on drywall sheets.

Tool: Drywall Saw (or Keyhole Saw) - For cutting openings for electrical boxes, plumbing, and fixtures.

Tool: Drywall Knives (6-inch, 10-inch, 12-inch) - Various sizes for applying joint compound to seams and corners.

Tool: Mud Pan or Hawk - To hold joint compound while working.

Tool: Sanding Sponges or Pole Sander - For smoothing joint compound after it dries.

Tool: Drywall Lifter - Especially useful for installing drywall on ceilings or when working solo. (Optional - Rent)

Tool: Paintbrushes (various sizes) - Including angled sash brushes for cutting in around trim and corners, and larger brushes for general application.

Tool: Paint Rollers (with appropriate nap) - For efficient coverage of large wall and ceiling surfaces.

Tool: Paint Trays or Roller Bucket - For holding paint and loading rollers.

Tool: Painter's Tape - To mask off trim, windows, and other areas you don't want painted.

Tool: Drop Cloths/Plastic Sheeting - To protect floors, fixtures, and furniture from paint drips and spills.

Tool: Spackle/Filler and Spackling Knife - For filling nail holes and minor imperfections in drywall before priming.

Tool: Sandpaper/Sanding Sponge - For smoothing patched areas and preparing surfaces for paint.

Tool: Extension Pole - For paint rollers, to comfortably reach ceilings and high walls. (Optional)

Tool: Ladder/Stepladder - For safe access to high areas.

Tool: Cleaning Supplies - Including a mild detergent and sponges for cleaning walls before priming/painting.

- New Porcelain Tile Flooring (tier 3 proceed)

1. Subfloor Preparation:

- * Ensure your subfloor is clean, dry, level, and structurally sound. Any movement or unevenness can lead to cracked tiles. Sweep and vacuum thoroughly.

- * Install cement backer board over the subfloor. Cut boards to size using a utility knife (scoring deeply and snapping) and secure them with specialized screws, ensuring seams are staggered. Leave a small gap (typically 1/8 inch) between boards and around the perimeter where they meet the wall.

- * Tape all seams of the backer board with fiberglass mesh tape and cover with a thin, even layer of thin-set mortar using a flat trowel. Allow this to dry.

2. Layout Planning (Dry Run):

- * Find the center of your room by snapping chalk lines between the midpoints of opposite walls.

- * Dry-lay tiles (without mortar) starting from the center outward, working in a section no larger than you can comfortably tile in a few hours. This allows you to plan your cuts and adjust for the best aesthetic, minimizing small, awkward cuts at visible edges. Aim for larger tile pieces in high-traffic or visible areas.

- * Adjust your starting point if necessary to ensure that cuts at perimeter walls are at least half a tile wide. Snap new chalk lines or draw reference lines on the backer board for your final starting point.

3. Mixing Thin-Set Mortar:

- * Mix thin-set mortar according to the manufacturer's instructions. It should have a smooth, peanut butter-like consistency. Mix small batches as needed, as it has a limited working time once mixed.

4. Applying Thin-Set and Laying Tiles:

- * Using a notched trowel (with the notch size recommended for your tile), spread a layer of thin-set over a small section of the floor, no larger than you can comfortably tile within 15-20 minutes. Hold the trowel at a 45-degree angle to create uniform ridges.

- * Gently place the first tile onto the mortar, aligning it with your reference lines. Wiggle it slightly and press down firmly to ensure good contact and collapse the mortar ridges. You can lift a tile occasionally to check for 90-100% thin-set coverage on the back.

- * Insert tile spacers between tiles to maintain consistent grout lines.

- * Continue laying tiles, working your way out from your starting point. Use a rubber mallet to gently tap tiles into place.

- * Continuously check for flatness and levelness using your 2-foot and 4-foot levels. Adjust tiles as you go.

- * Cut tiles as needed for edges and around obstacles using your wet tile saw.

5. Curing:

- * Allow the thin-set mortar to cure completely, typically 24-48 hours (or as specified by the manufacturer), before

walking on the tiles, removing spacers, or grouting.

6. Grouting:

- * Once the thin-set is fully cured, remove all tile spacers.
- * Mix grout according to manufacturer's instructions. It should be a consistent, workable paste.
- * Apply grout to the tile surface using a rubber float, pushing it firmly into all grout lines at a 45-degree angle. Ensure all gaps are filled.
- * Work in small sections (e.g., 10-20 sq ft at a time).
- * Wipe off excess grout from the tile surface with the rubber float, cleaning the float frequently.
- * After 15-30 minutes (or as recommended by the manufacturer, depending on grout type and humidity), use a damp (not soaking wet) sponge to carefully wipe the tiles clean, rinsing the sponge frequently in clean water. Be careful not to pull too much grout out of the lines.
- * Allow the grout to cure (usually 24-72 hours). A slight haze may appear on the tiles; this can be buffed off with a clean, dry cloth or a specialized grout haze remover after the grout has dried.

7. Sealing (if necessary):

- * Porcelain tile itself typically does not need sealing, but some grout types do. Check the manufacturer's recommendations for your specific grout. If sealing is recommended, apply a grout sealer as directed.

Tool: Safety Glasses and Gloves - Essential for protecting your eyes and hands during cutting and mixing. (Buy)

Tool: Dust Mask/Respirator - To protect your lungs from tile dust. (Buy)

Tool: Knee Pads - Highly recommended for comfort and protection during tiling. (Buy)

Tool: Wet Tile Saw - Absolutely essential for making clean, precise cuts on porcelain tile. (Consider buying a budget model, or renting a good quality one for the duration of the tiling work)

Tool: Notched Trowel - You'll need one with the correct notch size for your specific porcelain tiles and thin-set mortar. The tile manufacturer or thin-set instructions will specify the recommended notch size. (Buy)

Tool: Grout Float - For pressing grout into the tile joints. (Buy)

Tool: Tile Spacers - To ensure uniform grout lines between tiles. (Buy)

Tool: Large Sponges and Buckets - For cleaning excess grout from tiles and for mixing. (Buy)

Tool: Rubber Mallet - To gently tap tiles into place and ensure good adhesion. (Buy)

Tool: Level (2-foot and 4-foot) - Crucial for ensuring your floor tiles are perfectly level. (Buy)

Tool: Measuring Tape, Pencil, and Carpenter's Square - For accurate measurements and marking cuts. (Buy)

Tool: Mixing Drill and Paddle Attachment - For thoroughly mixing thin-set mortar and grout. If you have a powerful corded drill, you can buy just the paddle attachment. (Buy paddle, use existing drill or rent a dedicated mixer)

Tool: Cement Backer Board Screws - Specific screws designed for securing backer board to the subfloor. (Buy)

Tool: Fiberglass Mesh Tape - For sealing seams in backer board. (Buy)

- New Vanity, Countertop, and Undermount Sink (tier 3 proceed)

1. Prepare the Area:

Ensure the floor where the vanity will sit is clean, level, and free of debris. Ensure wall studs are accessible and clearly marked if the vanity needs to be attached to them.

Confirm that water supply lines and the drain pipe are in the correct location and accessible for connection to the new sink.

2. Assemble the Vanity Cabinet (if required):

Many vanities come flat-packed and require assembly. Follow the manufacturer's instructions carefully. Ensure all joints are tight and square.

3. Position and Level the Vanity:

Place the assembled vanity cabinet in its desired location against the wall.

Use a level to check if the vanity is plumb and level from front to back and side to side. Use wood shims underneath the cabinet if the floor is uneven. Tap shims gently into place and trim any excess with a utility knife.

Once level, use a stud finder to locate wall studs behind the vanity. Secure the vanity to the wall studs using

appropriate screws (e.g., 2.5-3 inch wood screws), driving them through the vanity's back panel or mounting strips. This prevents the vanity from tipping.

4. Install the Undermount Sink to the Countertop:

This step is typically done before placing the countertop on the vanity. Flip the countertop upside down on a protected, flat surface.

Carefully position the undermount sink over the opening on the underside of the countertop, ensuring it's centered and aligned.

Apply a continuous bead of silicone caulk or other recommended adhesive around the rim of the sink where it meets the countertop. Some sinks may also use clips; if so, install them now.

Firmly press the sink into place and secure it with clamps (if using) until the adhesive cures according to the manufacturer's instructions. Wipe away any excess caulk/adhesive immediately.

5. Install the Faucet and Drain Assembly (to the sink/countertop):

Before placing the countertop onto the vanity, install the faucet and drain assembly to the sink and/or countertop, following their respective manufacturer's instructions. It's much easier to do this while the countertop is not yet on the vanity, as access from below is often restricted once installed.

Ensure all connections are hand-tightened first, then gently snugged with a wrench. Do not overtighten.

6. Place the Countertop:

Once the sink, faucet, and drain assembly are installed (and any adhesives cured), carefully lift the countertop (with the sink attached) and place it onto the vanity cabinet. Ensure it's properly aligned with the cabinet edges and the wall.

Apply a thin bead of silicone caulk or construction adhesive along the top edges of the vanity cabinet before setting the countertop in place to secure it. Some countertops may use clips or screws from underneath.

7. Connect Plumbing:

Connect the hot and cold water supply lines from the wall to the faucet's supply lines. Use adjustable or basin wrenches, ensuring connections are tight but not overtightened. Use thread seal tape on threaded connections if recommended.

Connect the P-trap assembly from the sink's drain tailpiece to the wall drain pipe. Ensure all slip-joint connections are hand-tightened, then snugged with pliers, making sure washers are correctly seated to prevent leaks.

8. Test for Leaks:

Slowly turn on the water supply to the bathroom.

Run water through the faucet, filling the sink and letting it drain. Carefully inspect all plumbing connections (supply lines, drain, P-trap) for any leaks. Tighten connections as needed.

9. Final Sealing and Cleanup:

Apply a thin, continuous bead of silicone caulk along the seam where the back edge of the countertop meets the wall. This prevents water from seeping behind the vanity.

Wipe away any excess caulk or adhesive.

Install any drawers or doors on the vanity cabinet.

Tool: Safety Glasses and Gloves - Essential for protecting your eyes and hands during handling and installation.

Tool: Tape Measure and Pencil - For accurate measurements and marking during placement.

Tool: Level (2-foot or 4-foot) - Crucial for ensuring the vanity and countertop are perfectly level.

Tool: Stud Finder - To locate wall studs for secure mounting of the vanity.

Tool: Drill and Drill Bits - For pilot holes and driving screws to secure the vanity to the wall.

Tool: Screwdriver Set (Phillips and Flathead) - For various vanity components and securing hardware.

Tool: Adjustable Wrench or Basin Wrench - For connecting water supply lines and drain pipes to the sink. A basin wrench is particularly useful for tight spaces under the sink.

Tool: Pliers (Channel Lock or Tongue-and-Groove) - Useful for holding pipes or nuts during plumbing connections.

Tool: Caulking Gun and Silicone Caulk (Kitchen & Bath Grade) - For sealing the countertop to the vanity and against

the wall, and the sink to the countertop.

Tool: Utility Knife - For opening packaging, scoring caulk, or trimming excess.

Tool: Wood Shims - For leveling the vanity if your floor is uneven.

Tool: Clamps (Bar Clamps or F-Clamps) - To hold the undermount sink securely in place while the adhesive cures. (Optional, but recommended for undermount sinks)

Tool: Adhesive (Silicone or Epoxy) - Specifically recommended by your sink or countertop manufacturer for undermount sink installation.

- New Toilet (tier 3 proceed)

1. Preparation:

Ensure the old toilet is removed and the flange is clear of old wax or debris. The toilet flange should be securely fastened to the floor. Inspect the flange for any damage and replace it if necessary.

Have a new wax ring (with or without a plastic horn, depending on your flange type) ready, along with new closet bolts if your kit doesn't include them.

2. Install New Closet Bolts:

Insert the new closet bolts into the slots on the toilet flange. Position them directly across from each other (at 3 and 9 o'clock) and slide the plastic washers onto them to hold them upright.

3. Place the Wax Ring:

Carefully place the new wax ring onto the horn of the toilet's waste outlet (the bottom of the toilet) or directly over the toilet flange, ensuring it's centered. Do not touch the wax ring once placed, as handling can deform it and compromise the seal.

4. Set the Toilet:

Lift the new toilet bowl and carefully lower it onto the closet bolts, aligning the bolt holes with the bolts. Guide it down squarely.

Apply even downward pressure, gently rocking the toilet back and forth slightly to compress the wax ring. Do not twist the toilet, as this can break the wax seal.

Once the toilet is seated, ensure it is level. You may use shims (plastic ones are best for bathrooms) if necessary, but don't over-tighten the bolts to compensate for an uneven floor.

5. Secure the Toilet:

Place the plastic washers, then metal washers, and then nuts onto the closet bolts.

Hand-tighten the nuts first. Then, use a wrench to gently tighten the nuts, alternating between them, until the toilet is snug and stable on the floor. Be careful not to overtighten, as this can crack the porcelain.

Once tightened, snap the plastic caps over the bolts (you may need to trim the bolts with a hacksaw if they are too long).

6. Connect Water Supply:

Connect the new flexible water supply line to the toilet's fill valve and to the supply stop on the wall. Hand-tighten first, then use a wrench for a final snug turn (do not overtighten). Ensure washers are correctly seated for a watertight seal.

7. Turn on Water and Test:

Slowly turn the water supply back on at the stopcock or main valve.

Check for leaks around the supply line, the base of the toilet, and inside the tank. Allow the tank to fill completely.

Flush the toilet several times to ensure it fills and drains properly and to check for any additional leaks, especially around the base and connections.

Tool: Adjustable Wrench or Basin Wrench - For connecting the water supply line and tightening the toilet bolts. A basin wrench is particularly useful for tight spaces under the toilet tank. (Buy)

Tool: Pliers (Channel Lock or Tongue-and-Groove) - Useful for holding pipes or nuts. (Buy)

Tool: Screwdriver Set (Phillips and Flathead) - For various fasteners. (Buy)

Tool: Utility Knife - For opening packaging or trimming the wax ring if needed. (Buy)

Tool: Wax Ring with Flange - This seals the toilet to the waste pipe. Ensure you get one with a plastic horn. (Buy)

Tool: New Toilet Bolts (if not included with your new toilet) - These secure the toilet to the floor flange. (Buy)

Tool: Shims (plastic) - Useful for leveling the toilet if your floor is slightly uneven. (Buy)

- Tub Glass Separator (tier 3 proceed)

1. Preparation and Layout:

Ensure the tub deck, floor, and wall surfaces where the separator will be installed are clean, dry, level, and free of debris.

Unpack the glass panel(s) and hardware. Inspect for any damage. Handle glass carefully to avoid chipping edges.

Measure and mark the desired position for the bottom track (if applicable) on the tub deck and the vertical wall channel(s) on the wall. Use a level to ensure all lines are perfectly straight and plumb. For frameless panels, mark the hinge and clamp locations precisely.

Use a stud finder to identify wall studs if any part of the mounting hardware needs to be secured directly into wood for maximum support.

2. Install Mounting Hardware (Channels, Hinges, Clamps):

Bottom Track/Channel (if applicable): Apply a bead of silicone caulk to the underside of the bottom track and firmly press it onto the tub deck, aligning with your marks. Some systems may require drilling and screwing this down.

Wall Channel(s): Hold the vertical wall channel(s) against your marked lines on the wall. Mark the drill points. For tile, use a masonry or tile bit to drill pilot holes. Insert appropriate wall anchors (e.g., plastic sleeves or toggle bolts, depending on your wall material and stud location). Apply silicone caulk to the back of the channel before screwing it securely to the wall.

Hinges/Clamps (for frameless/pivot doors): Precisely mark and drill holes for hinges or clamps on the wall, ensuring they are perfectly aligned and level/plumb. Install the back plates or fixed halves of these components.

3. Install the Glass Panel(s):

This step often requires two people due to the weight and fragility of glass. Use glass suction cups for better grip and control.

Carefully lift the glass panel and guide it into the installed channels or attach it to the hinges/clamps. Ensure the panel is seated correctly and evenly.

Secure the glass panel within the channels or fasten the remaining parts of the hinges/clamps according to the manufacturer's instructions. Do not overtighten screws, as this can crack the glass.

Use your level to verify that the glass is perfectly plumb and level.

4. Seal the Unit:

Apply a continuous, even bead of mildew-resistant silicone caulk along all seams where the glass, channels, and hardware meet the tub and walls. This is crucial for preventing water leaks.

Smooth the caulk line with a wet finger or a caulking tool for a clean finish. Allow the caulk to cure completely (typically 24-48 hours) before using the shower.

5. Final Adjustments and Cleanup:

Install any handles, knobs, or magnetic strips for doors as per instructions.

Clean any fingerprints or smudges from the glass. Wipe away any excess caulk.

Tool: Safety Glasses and Gloves - Essential for protecting your eyes and hands, especially when handling glass.

Tool: Tape Measure and Pencil - For accurate measurements and marking for placement.

Tool: Level (2-foot or 4-foot) - Crucial for ensuring the glass panel and any mounting hardware are perfectly level and plumb.

Tool: Drill and Drill Bits (including masonry/tile bits if drilling into tile, and wood/metal bits) - For creating pilot holes and mounting hardware.

Tool: Stud Finder - To locate wall studs for secure mounting of any fixed panels or support bars.

Tool: Caulking Gun and Silicone Caulk (Kitchen & Bath Grade, mildew-resistant) - For sealing the glass panel and mounting channels against water penetration.

Tool: Utility Knife - For scoring caulk and trimming excess.

Tool: Screwdriver Set (Phillips and Flathead) - For various hardware components.

Tool: Rubber Mallet - To gently tap in anchors or assist with alignment.

Tool: Glass Suction Cups - For safely handling large, heavy glass panels. These can often be rented or purchased at hardware stores.

- Shower Niche (tier 3 proceed)

1. Plan Niche Location and Size:

- * Before closing up the walls with cement board, use a stud finder to locate the existing wall studs in your shower area.

- * Determine the ideal location and size for your niche. Niches are typically built *between* two existing studs, or you may need to cut a stud and add horizontal blocking (header and sill) for a wider niche. Avoid external walls if possible to prevent thermal bridging and potential condensation issues, though this is not always feasible.

- * Mark the exact dimensions of your desired niche opening on the studs.

2. Rough Framing (if necessary):

- * If building between two studs, you may just need to add horizontal blocking (niche header and sill) between them to define the top and bottom of the niche. Cut 2x4s to fit snugly between the studs at the desired height and secure them with screws or nails.

- * If you need a wider niche that spans more than one stud, you'll need to cut out a section of one or more studs and create a new framed opening with headers and cripple studs, similar to framing a window. This can be more complex and might involve minor structural considerations (though typically not Tier-1 for small niches). Ensure the new framing is secure.

3. Install Backer Board:

- * Install your cement backer board (or other tile-ready substrate) onto the wall studs, extending into the niche opening. Cut the backer board precisely around the framed niche opening.

- * Secure the backer board to the framing with specialized screws, ensuring all edges are flush and seams are tight.

4. Waterproof the Niche:

- * This is a critical step for preventing leaks. Apply a waterproofing membrane (sheet or liquid) over the entire shower area, paying special attention to the niche. Extend the membrane seamlessly into the niche and over all edges. Overlap seams according to the manufacturer's instructions.

- * For liquid membranes, apply multiple coats as directed. For sheet membranes, ensure corners and seams are properly sealed with the manufacturer's band or sealant.

- * Use waterproof caulk or sealant at all internal corners of the niche where the backer board meets. Ensure the niche is fully sealed and watertight.

5. Tile the Niche Interior:

- * Start by tiling the back wall of the niche, then the top, bottom, and finally the side walls. This ensures any cut edges are less visible.

- * Mix thin-set mortar and apply it to the backer board within the niche using a small notched trowel. Apply a thin layer to the back of each tile (back-buttering) for maximum adhesion.

- * Carefully set your tiles, using tile spacers for consistent grout lines. Ensure the bottom of the niche is slightly sloped forward (towards the shower opening) to allow water to drain out, preventing pooling.

- * Allow the thin-set to cure completely.

6. Grout the Niche:

- * Mix and apply grout to the niche, pressing it firmly into all grout lines with a grout float. Wipe off excess grout with a damp sponge.

- * Allow the grout to cure.

7. Caulking/Sealing:

- * Apply a thin bead of 100% silicone caulk (mildew-resistant) at all changes of plane within the niche (e.g., where the

back meets the sides, and the sides meet the top/bottom). Grout can crack in these areas, so caulk provides a flexible, watertight seal.

Tool: Safety Glasses and Gloves - Essential for protecting your eyes and hands, especially when cutting materials.

Tool: Tape Measure and Pencil - For accurate measurements and marking during framing and niche placement.

Tool: Level (2-foot) - Crucial for ensuring the niche is perfectly level and plumb.

Tool: Stud Finder - Essential for locating existing wall studs to determine optimal niche placement and for securing any new framing.

Tool: Utility Knife - For cutting cement backer board and waterproofing membranes.

Tool: Drywall Saw or Jigsaw - If you need to cut into existing drywall to expose studs.

Tool: Drill and Drill Bits - For pilot holes and driving screws to secure new framing and backer board.

Tool: Wood Saw (Hand Saw or Circular Saw) - For cutting wood studs for niche framing.

Tool: Cement Board Scoring Tool or Angle Grinder with Diamond Blade - For precise, clean cuts on cement backer board if not using a utility knife. (Buy or Rent if you prefer the grinder)

Tool: Caulking Gun and Waterproof Sealant/Adhesive - For sealing seams and corners within the niche during waterproofing.

Tool: Trowel (flat and notched) - For applying thin-set mortar.

Tool: Grout Float and Sponges - For tiling the niche interior.

Tool: Tile Spacers - For consistent grout lines within the niche.

Tool: Mixing Drill and Paddle Attachment - For mixing thin-set mortar. (Buy paddle, use existing drill)

- New Light Fixtures (tier 3 proceed)

1. Safety First - Turn Off Power:

Crucially, turn off the electrical power to the bathroom circuit at your main electrical panel. This is the most important step to prevent electrical shock.

Use your non-contact voltage tester to confirm that the power is off at the light switch and at the fixture itself before touching any wires.

2. Remove Old Fixture:

Carefully remove any glass shades, bulbs, or decorative elements from the old fixture.

Unscrew the mounting nuts or screws that hold the fixture to the electrical box.

Gently lower the fixture to expose the wiring connections.

Take note of how the wires are connected (e.g., black to black, white to white, ground to ground). Taking a photo with your phone can be helpful.

Disconnect the wire nuts and separate the wires. Keep the old fixture aside for disposal.

3. Prepare New Fixture for Installation:

Unpack your new light fixture and review the manufacturer's instructions. Assemble any necessary components according to the instructions.

Attach the new mounting bracket (if provided and different from the old one) to the electrical box in the ceiling or wall. Ensure it is secure.

4. Wire the New Fixture:

Carefully strip about 1/2 to 3/4 inch of insulation from the ends of the wires on both the new fixture and from the electrical box, if they are not already stripped.

Connect the wires: Typically, the black wire from the fixture connects to the black wire (hot) from the electrical box. The white wire from the fixture connects to the white wire (neutral) from the electrical box. The bare copper or green wire from the fixture connects to the bare copper or green ground wire from the electrical box, and also to the ground screw on the mounting bracket or electrical box if there is one.

Twist the wire ends together clockwise and secure them firmly with new wire nuts. Gently tug on each wire connection to ensure it is secure. Cover any exposed wire with electrical tape if desired.

5. Mount the New Fixture:

Carefully tuck the connected wires into the electrical box.

Align the new fixture with the mounting bracket and secure it in place with the provided screws or nuts. Be careful not to pinch any wires.

6. Install Bulbs and Shades:

Install the recommended light bulbs into your new fixture.

Attach any glass shades or decorative covers according to the manufacturer's instructions.

7. Restore Power and Test:

Go back to your main electrical panel and turn the power back on to the bathroom circuit.

Test the new light fixture(s) to ensure they operate correctly.

Tool: Safety Glasses - Always essential for protecting your eyes during electrical work.

Tool: Non-Contact Voltage Tester - For ensuring power is off before working with electrical wires.

Tool: Ladder or Step Stool - For safe access to ceiling or high wall fixtures.

Tool: Screwdriver Set (Phillips and Flathead) - For removing old fixtures, assembling new ones, and securing cover plates.

Tool: Wire Strippers/Cutters - For preparing electrical wires.

Tool: Pliers (Needle-nose or Lineman's) - For bending and manipulating wires, especially in tight spaces.

Tool: Wire Nuts - For making secure electrical connections (often included with new fixtures).

Tool: Electrical Tape - For securing connections or labeling wires, if needed.

Tool: Utility Knife - For opening packaging or trimming insulation.

Tool: Pencil and Tape Measure - For marking and accurate positioning, especially for vanity lights or multiple fixtures.

- Wall-Mounted Mirror, Freestanding Cabinet, and Wall Hanger (tier 3 proceed)

1. Phase 1: Installing the Wall-Mounted Mirror

2. Determine Placement:

3. Decide on the exact height and centering for your mirror, typically above the vanity. Consider the height of those who will use it most.

4. Use a tape measure and pencil to mark the desired top and center points on the wall.

5. Locate Mounting Points:

6. Consult your mirror's hanging instructions. Some mirrors use a French cleat, some use D-rings, and others might have keyhole hangers or direct screw-through brackets.

7. Measure the distance between the mounting points on the back of the mirror. Transfer these measurements accurately to your wall, ensuring they are perfectly level using your level.

8. Use a stud finder to see if your mounting points align with wall studs. Mounting directly into studs is always the most secure option.

9. Install Wall Anchors/Screws:

10. If mounting into studs, drill pilot holes and drive screws directly into the studs, leaving enough of the screw head exposed to engage with the mirror's hanging mechanism.

11. If mounting into drywall without a stud, drill appropriate pilot holes and install suitable wall anchors (e.g., toggle bolts for heavier mirrors, self-drilling drywall anchors for lighter ones). Then, drive screws into the anchors.

12. Ensure all anchors/screws are securely installed and can support the weight of the mirror.

13. Hang the Mirror:

14. Carefully lift the mirror and align its hanging mechanism with the installed screws or hangers on the wall.

15. Gently lower the mirror until it is securely seated.

16. Step back and check that the mirror is level. Adjust if necessary.

17. Final Check: Give the mirror a gentle tug to ensure it's firmly attached to the wall.

18. Phase 2: Assembling and Positioning the Freestanding Cabinet

19. Assemble the Cabinet:

20. Follow the manufacturer's instructions to assemble your freestanding cabinet. Ensure all joints are tight and

square.

21. Choose Location:

22. Place the assembled cabinet in its desired location, ensuring it does not obstruct traffic flow, door swings, or access to other fixtures.

23. Level the Cabinet:

24. Use a level to check if the cabinet is level. Many freestanding cabinets have adjustable feet; use these to level the cabinet if your floor is uneven.

25. Secure to Wall (Optional, but Recommended for Safety):

26. For safety, especially in homes with children or pets, it's highly recommended to secure freestanding furniture to the wall to prevent tipping.

27. Use a stud finder to locate a wall stud behind the top of the cabinet.

28. Attach the anti-tip hardware (usually included with the cabinet) to both the cabinet and the wall stud according to the manufacturer's instructions.

29. Phase 3: Installing the Wall Hanger (for Clothes and Towels)

30. Determine Placement:

31. Decide on the optimal height and location for your wall hanger, considering ease of access and clearance from other fixtures.

32. Use a tape measure and pencil to mark the desired mounting points on the wall.

33. Locate Mounting Points:

34. As with the mirror, use a stud finder to check if your mounting points align with wall studs. Towel bars or robe hooks often hold damp towels, so mounting into a stud is ideal for long-term stability.

35. Install Wall Anchors/Screws:

36. If mounting into studs, drill pilot holes and drive screws directly.

37. If mounting into drywall without a stud, use appropriate wall anchors that can support the weight of the hanger plus the items it will hold (e.g., a wet towel can be surprisingly heavy).

38. If mounting into tile, you will need a specialized tile drill bit. Start with a smaller pilot hole, then use the recommended size for your anchor. Install tile anchors or plastic sleeves into the drilled holes.

39. Attach the Hanger:

40. Secure the wall hanger to the installed screws or anchors according to the manufacturer's instructions.

41. Ensure it is firmly attached and level.

Tool: Tape Measure and Pencil - Essential for accurate measurements and marking placement points on the wall. (Buy)

Tool: Level (2-foot or 4-foot) - Crucial for ensuring the mirror, cabinet, and hanger are perfectly level. (Buy)

Tool: Stud Finder - Recommended for locating wall studs, especially for mounting heavier items like large mirrors or if the wall hanger will bear significant weight. (Buy)

Tool: Drill and Drill Bits (appropriate for wall material - wood, drywall, or tile) - For drilling pilot holes and securing mounting hardware. (Buy)

Tool: Screwdriver Set (Phillips and Flathead) - For various hardware components and securing screws. (Buy)

Tool: Wall Anchors - Depending on the weight of the item and your wall material (drywall, plaster), you may need specific anchors (e.g., drywall anchors, toggle bolts). (Buy)

Tool: Utility Knife - For opening packaging. (Buy)

Tool: Blue Painter's Tape - Can be used to mark drill spots or to temporarily hold templates in place. (Optional)

Tool: Safety Glasses - Recommended when drilling. (Buy)

Tool: Heavy-Duty Gloves - Protect your hands from cuts, scrapes, and chemicals.

Tool: Safety Glasses/Goggles - Crucial for protecting your eyes from flying debris and dust.

Tool: Dust Mask/Respirator - Essential for protecting your lungs from dust, especially during demolition and tile work.

Tool: Sturdy Closed-Toe Shoes/Work Boots - Protect your feet from falling objects and nails.

Tool: Pry Bar Set - For removing trim, baseboards, and breaking up drywall.

Tool: Claw Hammer - A basic hammer is useful for various tasks.

Tool: Utility Knife - For scoring drywall and cutting various materials.

Tool: Large Buckets/Trash Cans - For collecting debris as you go.

Tool: Heavy-Duty Trash Bags - For bagging up smaller debris.

Tool: Shovel and Broom - For cleanup.

Tool: Wheelbarrow or Heavy-Duty Dolly - For moving heavier debris, like old toilets or large sections of tile.

Tool: Demolition Hammer / Jackhammer - If your old tile is set in a thick mortar bed, a demo hammer will save a lot of time and effort. You can rent a smaller electric demolition hammer for approximately \$48 to \$89 per day, or \$192 to \$139 per week.

Tool: Wet Tile Saw - Absolutely essential for making clean, precise cuts on porcelain tile.

Tool: Notched Trowel - You'll need one with the correct notch size for your specific porcelain tiles and thin-set mortar.

Tool: Grout Float - For pressing grout into the tile joints.

Tool: Tile Spacers - To ensure uniform grout lines between tiles.

Tool: Large Sponges and Buckets - For cleaning excess grout from tiles.

Tool: Rubber Mallet - To gently tap tiles into place and ensure good adhesion.

Tool: Level (2-foot and 4-foot) - Crucial for ensuring your floor and wall tiles are perfectly level and plumb.

Tool: Measuring Tape, Pencil, and Carpenter's Square - For accurate measurements and marking cuts.

Tool: Mixing Drill and Paddle Attachment - For thoroughly mixing thin-set mortar and grout. If you have a powerful corded drill, you can buy just the paddle attachment.

Tool: Cement Backer Board Screws - Specific screws designed for securing backer board.

Tool: Fiberglass Mesh Tape - For sealing seams in backer board.

Tool: Waterproof Membrane/Liquid - For the tub surround, as discussed in the procedures.

Tool: Caulking Gun and Silicone Caulk (Kitchen & Bath Grade) - For sealing corners and edges where tiles meet other surfaces, especially in the shower.

Tool: Adjustable Wrench or Basin Wrench - For connecting the water supply line and tightening the toilet bolts. A basin wrench is particularly useful for tight spaces under the toilet tank.

Tool: Pliers (Channel Lock or Tongue-and-Groove) - Useful for holding pipes or nuts.

Tool: Screwdriver Set (Phillips and Flathead) - For various fasteners.

Tool: Wax Ring with Flange - This seals the toilet to the waste pipe. Ensure you get one with a plastic horn.

Tool: New Toilet Bolts - These secure the toilet to the floor flange.

Tool: Shims (plastic) - Useful for leveling the toilet if your floor is slightly uneven.

6b. Add to Your Contractor's Scope

Eligible work you chose to hand to a professional:

- New Exhaust Fan (tier 2 permitted)

7. Advisory Checklist

- Itemized Bid
- Apples-to-Apples Comparison
- Verify Credentials
- "Invisible" Inclusions: Demolition and Debris Removal/Haul
- "Invisible" Inclusions: Waterproofing
- "Invisible" Inclusions: Rough-in Plumbing/Electrical
- "Invisible" Inclusions: Permit Fees
- "Invisible" Inclusions: Final Cleanup
- Payment Schedule

- Warranty Terms
- Change-Order Procedure
- Project Timeline
- Budget Expectation

8. Phase Checklists

Before demolition:

- Order long-lead materials first.
- Confirm permits filed.

After demolition:

- Inspect for hidden conditions before closing walls.

While in progress:

- Keep waterproofing before tile.
- Hit inspection milestones.

Wrap-up:

- Final inspection sign-off before final payment.

9. Budget Gap - Your On-Ramp

Gap to bridge: \$3,000. This is a plan on-ramp, not a dead end.

- Phase the work over time as budget allows.
- Self-perform eligible non-professional items to reduce labor.
- Shift finishes toward economy bands without changing the layout.